

We consider how to initialize the parameters of a model before we start training.

1. Why is proper initialization important?

During forward process:

$$\begin{aligned} f_k &= \beta_k + \Omega_k a_k \\ &= \beta_k + \Omega_k \alpha(f_{k-1}) \end{aligned}$$

→ activation function, e.g. ReLU

if β_k is initialized to 0 and Ω_k is initialized such that each element follows $N(0, \sigma^2)$ independently.

Then if σ is small, e.g. 0.001, then the magnitude of $f_k \ll f_{k+1}$

if σ is large, e.g. 100, then the magnitude of $f_k \gg f_{k+1}$

in deep neural networks, this is even more serious, because the magnitude problem will be worse and worse through each layer.

During backpropagation:

$$\begin{aligned} \frac{\partial L}{\partial f_{k-1}} &= \frac{\partial L}{\partial f_k} \frac{\partial f_k}{\partial f_{k-1}} \\ &= \frac{\partial L}{\partial f_k} \Omega_k \frac{\partial (\alpha(f_{k-1}))}{\partial f_{k-1}} \\ &= \frac{\partial L}{\partial f_k} \Omega_k \text{diag}(\mathbf{1}(f_{k-1} > 0)) \end{aligned}$$

(if $\alpha = \text{relu}$)

depends on the scale of Ω_k , there is risk (if we do not initialize properly) that the gradients

- getting smaller and smaller
- getting longer and longer

hard to train!

Solution: we want to keep the variance unchanged during both

forward pass: $\text{Var}(f_k) \approx \text{Var}(f_{k-1})$

backward propagation: $\text{Var}(\frac{\partial L}{\partial f_{k-1}}) \approx \text{Var}(\frac{\partial L}{\partial f_k})$

Derivation: β_k will be initialized to 0

1. forward pass:

notations: $f: f_{k-1}$
 $f': f_k$
 $f_i: i$ th element of f
 $f'_i: i$ th element of f'

$\beta: \beta_k$
 $\Omega: \Omega_k$

$$E(f'_i) = E(\beta_i + \sum_{j=1}^m \Omega_{ij} a(f_j))$$

$$= E(\beta_i) + E(\sum_{j=1}^m \Omega_{ij} a(f_j))$$

$$= 0 + \sum_{j=1}^m E(\Omega_{ij} a(f_j))$$

$$= \sum_{j=1}^m E(\Omega_{ij}) E(a(f_j))$$

the parameters are initialized independently w.r.t the layer outputs

$$= \sum_{j=1}^m 0 \cdot E(a(f_j))$$

$$= 0$$

$$\text{Var}(f'_i) = \text{Var}(\beta_i + \sum_{j=1}^m \Omega_{ij} a(f_j))$$

$$= \text{Var}(\sum_{j=1}^m \Omega_{ij} a(f_j))$$

$$= \sum_{j=1}^m \text{Var}(\Omega_{ij}) \text{Var}(a(f_j))$$

$$= \sum_{j=1}^m \sigma_{\Omega}^2 (E(a(f_j)^2) - E(a(f_j))^2)$$

$$= \sum_{j=1}^m \sigma_{\Omega}^2 E(a(f_j)^2)$$

assume here $\alpha = \text{Relu}$

$$= \sigma_{\Omega}^2 \sum_{j=1}^m (P(f_j > 0) E(a(f_j)^2 | f_j > 0) + P(f_j \leq 0) E(a(f_j)^2 | f_j \leq 0))$$

total expectation

$$= \sigma_{\Omega}^2 \sum_{j=1}^m \frac{1}{2} E(f_j^2 | f_j > 0)$$

assume the input is symmetric around 0
 (this is should be reasonable if no preprocessing each input to standard normal)

$$= \sigma_{\Omega}^2 \sum_{j=1}^m \frac{1}{2} \sigma_f^2$$

use the symmetric assumption again

$$= \frac{1}{2} m \sigma_{\Omega}^2 \sigma_f^2$$

$E(f_j^2) = \sigma_f^2 = P(f_j > 0) E(f_j^2 | f_j > 0) + P(f_j \leq 0) E(f_j^2 | f_j \leq 0)$
 $= \frac{1}{2} \sigma_f^2 + \frac{1}{2} \sigma_f^2 = \sigma_f^2$

Conclusion: if we want: $\sigma_{f_i}^2 = \sigma_{f_j}^2$, we need to set

$$\sigma_{\Omega}^2 = \frac{2}{m} \quad (\text{He initialization})$$

↘ number of neurons for last layer, i.e. rows for Ω_{k-1}

2. Backward propagation: (follows the same logic)

First, recall that:

$$\begin{aligned} \frac{\partial L}{\partial f_{k-1}} &= \frac{\partial L}{\partial f_k} \frac{\partial f_k}{\partial f_{k-1}} \\ &= \frac{\partial L}{\partial f_k} \Omega_k \frac{\partial a(f_{k-1})}{\partial f_{k-1}} \\ &= \frac{\partial L}{\partial f_k} \Omega_k \text{diag}(I(f_{k-1} > 0)) \end{aligned}$$

for notation simplicity, let us denote $\frac{\partial L}{\partial f_k} = g'$ and $\frac{\partial L}{\partial f_{k-1}} = g$
 $\Omega_k = \Omega$, $f = f_{k-1}$ and $f' = f_k$

$$\begin{aligned} \text{So we have:} \quad E(g_i) &= E\left(\sum_{j=1}^{m'} g'_j \Omega_{ji} (I(f_j > 0))\right) \\ &= \sum_{j=1}^{m'} E(g'_j \Omega_{ji} I(f_j > 0)) \\ &= \sum_{j=1}^{m'} \Omega_{ji} E(g'_j I(f_j > 0)) \\ &= \sum_{j=1}^{m'} 0 E(g'_j I(f_j > 0)) \\ &= 0 \\ \text{Var}(g_i) = \sigma_g^2 &= \text{Var}\left(\sum_{j=1}^{m'} g'_j \Omega_{ji} (I(f_j > 0))\right) \\ &= E\left(\left(\sum_{j=1}^{m'} g'_j \Omega_{ji} I(f_j > 0)\right)^2\right) \\ &= E\left(\sum_{j=1}^{m'} (g'_j \Omega_{ji} I(f_j > 0))^2\right) + \sum_{j_1 \neq j_2} E(2 g'_{j_1} \Omega_{j_1 i} I(f_{j_1} > 0) g'_{j_2} \Omega_{j_2 i} I(f_{j_2} > 0)) \\ &= \sum_{j=1}^{m'} E(g_j^2 \Omega_{ji}^2 I(f_j > 0)^2) \\ &= \sum_{j=1}^{m'} E(g_j^2) E(\Omega_{ji}^2) E(I(f_j > 0)^2) \quad \text{assuming independence here} \\ &= m' \sigma_{g_j}^2 \sigma_{\Omega}^2 \frac{1}{2} \end{aligned}$$

Conclusion: if we want $\sigma_{g_i}^2 = \sigma_{g_j}^2$, we need $\sigma_{\Omega}^2 = \frac{2}{m'}$

If $m = m'$, then we can initialize Ω to keep both forward and backward stable

if $m \neq m'$, we use $\sigma_{\Omega}^2 = \frac{4}{m+m'}$ → replace m or m' with $\frac{m+m'}{2}$